



PRACTICAL QUESTION (SIMPLE VERSION)

Create an Angular application that:

1. Displays a list of students using `*ngFor`
2. Adds a student using **two-way data binding**
3. Shows/hides the list using `*ngIf`
4. Uses a **Service**
5. Uses `@Input()` and `@Output()`



STEP 1: Install Angular CLI (ONLY FIRST TIME)

Open VS Code → **Terminal** and type:

```
npm install -g @angular/cli
```

Check:

```
ng version
```



STEP 2: Create New Project

```
ng new student-app
```

```
cd student-app
```

```
write code .
```

Run:

```
ng serve
```

Open browser:

```
http://localhost:4200
```



STEP 3: What is Inside Angular Project? (VERY SIMPLE)

When you create project using:

```
ng new student-app
```

Angular creates many files.

You only need to remember **ONE FOLDER**:

📁 **src** → **app**

Open in VS Code:

```
student-app
```

```
├── src
```

```
│   └── app
```

Inside **app** folder, important files are:

File	Meaning
app.component.ts	Brain (logic)
app.component.html	Screen (design)
app.module.ts	Connection file

👉 Angular app runs using these files

STEP 4: Create Service (For Storing Names)

In VS Code terminal type:

```
ng generate service student
```

Now you will see a new file:

```
src/app/student.service.ts
```

STEP 5: Write Code in student.service.ts

Open file:

 student.service.ts

Delete everything and paste this:

```
import { Injectable } from '@angular/core';
```

```
@Injectable({
  providedIn: 'root'
})
export class StudentService {

  students: string[] = ['Aijaz', 'Rahul'];

  getStudents() {
    return this.students;
  }

  addStudent(name: string) {
    this.students.push(name);
  }
}
```



Meaning:

This file stores student names.

STEP 6: Open Main File (app.component.ts)

Path:

```
src/app/app.component.ts
```

Replace **FULL CODE** with this:

```
import { Component, OnInit } from '@angular/core';
import { StudentService } from '../student.service';
```

```
@Component({
  selector: 'app-root',
  templateUrl: './app.component.html'
})
export class AppComponent implements OnInit {

  students: string[] = [];
  show: boolean = true;
  newStudent: string = '';
```

```

constructor(private studentService: StudentService) {}

ngOnInit() {
  this.students = this.studentService.getStudents();
}

addStudent() {
  this.studentService.addStudent(this.newStudent);
  this.newStudent = '';
}
}

```

 **Meaning:**

This file controls the app.

STEP 7: Open Screen File (app.component.html)

Path:

src/app/app.component.html

Delete everything and paste:

```

<h2>Student App</h2>

<input [(ngModel)]="newStudent" placeholder="Enter name">
<button (click)="addStudent()">Add</button>

<br><br>

<button (click)="show = !show">Show / Hide List</button>

<ul *ngIf="show">
  <li *ngFor="let s of students">
    {{ s }}
  </li>
</ul>

```

 **Meaning:**

This file shows output on screen.

STEP 8: VERY IMPORTANT STEP (Fix Error)

Open file:

src/app/app.module.ts

Find this line at top:

```
import { BrowserModule } from '@angular/platform-browser';
```

Add this **below** it:

```
import { FormsModule } from '@angular/forms';
```

Now find:

```
imports: [
  BrowserModule
],
```

Change to:

```
imports: [
  BrowserModule,
  FormsModule
]
```

1,

👉 Now save all files



STEP 9: Run Project

In terminal:

ng serve

Open browser:

http://localhost:4200



WHAT YOU WILL SEE?

- ✓ A box to type name
 - ✓ Add button
 - ✓ Show/Hide button
 - ✓ List of students
-



NOW UNDERSTAND IN VERY SIMPLE WORDS

Thing	Meaning
ngModel	Connects textbox to variable
*ngFor	Shows list
*ngIf	Show / Hide
Service	Stores names
Component	Controls screen

REFERENCE PURPOSE:



STEP-BY-STEP (SCREENSHOT STYLE IN WORDS)



STEP 1: Open VS Code

1. Click **Start Menu**
2. Open **Visual Studio Code**

You will see an **empty window**.

● STEP 2: Open Terminal Inside VS Code

1. At the **top menu**, click:

Terminal → New Terminal

2. A **black screen** opens at bottom.
-

● STEP 3: Create Angular Project

In terminal, type:

```
ng new student-app
```

Then press **Enter**

● What You Will See:

It will ask:

Would you like to add Angular routing? (y/n)

👉 Type: **n** and press **Enter**

Which stylesheet format?

👉 Press **Enter** (default)

⌚ It will take 1–2 minutes.

● STEP 4: Open Project in VS Code

After finished, type:

```
cd student-app
```

```
code .
```

VS Code will **reload**.

● STEP 5: LOOK AT LEFT SIDE (THIS IS IMPORTANT)

On **LEFT SIDE PANEL**, you will see:

```
student-app
├── node_modules
├── src
├── angular.json
└── package.json
```

👉 Click the ► **arrow** near **src**

● STEP 6: Open src → app

Click like this:

➡ Click **src**

➡ Click **app**

Now you will see:

```
app
```

```
├── app.component.ts
├── app.component.html
└── app.module.ts
```

THESE FILES ARE ALREADY CREATED
You **DO NOT** create them.

STEP 7: Open Files (Like Screenshot Click)

Now click one by one:

- ✓ Click: `app.component.ts`
- ✓ Click: `app.component.html`
- ✓ Click: `app.module.ts`

They will open in **center screen**.

STEP 8: Create Service (Automatically)

Now in terminal (bottom), type:

```
ng generate service student
```

What You Will See on Left?

A new file appears:

```
src/app/student.service.ts
```

👉 Angular created it automatically.

STEP 9: Click This New File

On left:

- src
- app
- click **student.service.ts**

Now it opens in center.

VERY IMPORTANT RULE

You **NEVER** create files manually

You only:

- Click files
 - Delete code
 - Paste new code
-



VISUAL IMAGINE (TEXT DIAGRAM)

[VS CODE]

LEFT SIDE:

```
student-app
├── src
│   └── app
```

```
├── app.component.ts ← you click this
├── app.component.html ← you click this
├── app.module.ts
└── student.service.ts ← auto created
```

CENTER:

```
app.component.ts
(code here)
```